

Deposit Insurance and Depositor Behavior: Evidence from Colombia

Nicolás de Roux

Universidad de los Andes, Colombia

Nicola Limodio

Bocconi University, BAFFI CAREFIN, IGIER and CEPR, Italy

This paper studies the effect of deposit insurance on depositor behavior. Our theoretical framework integrates insights from public and financial economics and predicts that (1) deposit insurance induces bunching at the threshold in the deposit distribution and (2) an increase in the insurance threshold promotes deposit growth, particularly higher for individuals bunching at the initial limit. We exploit a large and unexpected increase in the Colombian insurance together with monthly depositor-level records from a major bank to test these predictions. We validate the existence of bunching in deposits and quantify the heterogeneous effect of deposit insurance on individual deposit growth. (*JEL* G21, G28, G51)

Received May 26, 2021; editorial decision October 24, 2022 by Editor Itay Goldstein. Authors have furnished an Internet Appendix, which is available on the Oxford University Press Web site next to the link to the final published paper online.

Financial regulation is central to promoting the resilience of financial institutions, avoiding bank failures, and setting the appropriate incentives for risk-taking. Among the various regulation tools, deposit insurance is one of

We are especially grateful to the editor, Itay Goldstein, and to two anonymous referees whose comments helped us to substantially improve the paper. We thank for their useful suggestions Isha Agarwal, Giorgia Barboni, Patrick Bolton, Olivier Darmouni, Filippo De Marco, Douglas W. Diamond, Carlo A. Favero, Kim Fe Cramer, Nicola Gennaioli, Giulia Giupponi, Juan Herreño, Sean Higgins, Marc Hofstetter, Agustín Hurtado, Monica Martinez-Bravo, Luis R. Martínez, Adrien Matray, Sergio Mayordomo, Nuno Palma, Krisztina Orban, Daniel Paravisini, Nicola Pavanini, Claudia Robles-Garcia, Raghuram Rajan, Rafael Repullo, Stefano Rossi, Nicolas Serrano-Velarde, Pablo Slutsky, Javier Suarez, Amir Sufi, and seminar participants at Bank of Spain, Bocconi University, CEMFI, Economic Research Southern Africa, the 3rd Endless Summer Conference on Financial Intermediation and Corporate Finance, KU Leuven, LUISS, Robert H. Smith School of Business at the University of Maryland, the 2021 Royal Economic Society Conference, Toulouse School of Economics, University of Manchester, and the WEFIDEV webinar on Finance and Development. We are grateful for the support of the Macro Finance Research Program at the University of Chicago and of Edward R. Allen. For funding, de Roux is grateful to Universidad de los Andes and Limodio to Bocconi University. We thank Riccardo Di Cato, Sergio Andrés Flórez-Orrego, Natalia Moreno, and Jacopo Tozzo for excellent research assistance. We are responsible for all errors. Supplementary data can be found on *The Review of Financial Studies* web site. Send correspondence to Nicola Limodio, nicola.limodio@unibocconi.it.

The Review of Financial Studies 36 (2023) 2721–2755

© The Author(s) 2022. Published by Oxford University Press on behalf of The Society for Financial Studies. All rights reserved. For permissions, please e-mail: journals.permissions@oup.com.

<https://doi.org/10.1093/rfs/hhac092>

Advance Access publication December 14, 2022

the most prominent (Diamond and Dybvig 1983; Diamond and Rajan 2000; Repullo 2000; Goldstein and Pautzner 2005; Bruche and Suarez 2010; Davila and Goldstein 2021) and has been extensively used in history, around the world, and during the 2008 financial crisis (Demirgüç-Kunt, Kane, and Laeven 2015; Martin, Puri, and Ufieri 2018). Nonetheless, this regulation creates costs since it encourages the moral hazard behavior of banks and increases the risk-taking of the financial system as a whole (Bryant 1980; Demirgüç-Kunt and Detragiache 2002; Cooper and Ross 2002; Calomiris and Jaremski 2019). On the other hand, it can have important benefits in terms of deposit inflows and financial stability. These costs and benefits should be weighted to determine the optimal level of deposit insurance (Allen et al. 2018; Davila and Goldstein 2021; van der Kwaak, Madeira, and Palma 2021). Therefore understanding the channels through which this specific regulation encourages deposits and estimating its causal effect on depositor behavior is critical.

Despite the relevance of this issue, few empirical studies have investigated how depositor behavior responds to deposit insurance, which mechanisms are at play, and the size of the effects. This knowledge gap develops from two constraints. First, we acknowledge important data limitations because access to high-frequency information at the depositor level is restricted and not accessible through publicly available sources. Second, and perhaps more importantly, deposit insurance typically changes in periods of financial turbulence, and the effect of the policy change cannot be disentangled from the effects of other concurring economic forces. For example, in times of financial distress, individuals may change their deposit behavior for reasons unrelated to changes in the insurance threshold. As a result, we lack stylized facts on the relation between depositor behavior and deposit insurance.

This paper offers a simple model and evidence that highlights a channel through which deposit insurance stimulates the size of banks and financial institutions: by promoting the safety of banks, it increases the expected deposit remuneration and attracts deposits into formal financial institutions. The model is a stylized theoretical framework that combines insights from recent papers on deposit insurance (Davila and Goldstein 2021) and elements from the modern public economics literature (Chetty et al. 2011; Kleven 2016). In this setting, the regulation shapes the distribution of bank deposits and induces a heterogeneous response of deposit growth to exogenous changes in the insurance threshold. We test the predictions of the model with event study and difference-in-differences (DiD) designs using depositor-level data at a monthly frequency from a large Colombian bank and a large and unexpected increase in the Colombian insurance threshold. Finally, we quantify the causal effect of deposit insurance on depositor behavior by obtaining an elasticity of deposit growth to deposit insurance.

The key elements of our model are a positive probability of bank default and a limited level of deposit insurance. Together, they create a convex kink in the expected deposit return, as its slope flattens for deposit levels above

the insurance threshold. This generates two intertwined predictions. First, the deposit distribution should exhibit “bunching”: the threshold of deposit insurance leads an excessive mass of depositors to hold the maximum amount of deposits insurable. Second, an exogenous increase in the threshold increases the returns of deposits and leads depositors to change the optimal allocation of their endowment in favor of higher deposits. Furthermore, this response is heterogeneous and depends on deposit holdings before the increase: individuals with balances at the previous threshold, the so-called “bunchers”, exhibit the strongest deposit response; those with amounts between the two thresholds show a moderate deposit growth, and those above the new threshold present the smallest increase.

To empirically investigate these predictions, we need a high-frequency and granular data set together with variation in a deposit insurance threshold unrelated to economic conditions. Therefore, we partnered with a large bank in Colombia and obtained 2 years of monthly records of individual deposit holdings for more than 50,000 individuals. Furthermore, we leverage a change in the deposit insurance threshold, allowing us to study the effect of deposit insurance on depositors in the absence of economic crises and bank runs. On April 18, 2017, the Colombian deposit guarantee fund (FOGAFIN, short for Fondo de Garantías de Instituciones Financieras) increased the insurance for individual deposits from 20 million Colombian pesos (COP) (approximately USD\$6,780) to 50 million COP (around USD\$16,950). The change was unexpected, and the announcement to the public occurred only one day after its implementation. Contrary to most changes in deposit insurance that are deliberated during financial crises, this increase aimed to update the real value of the insurance threshold, which had not changed since the year 2000.¹

Our contribution to the empirical identification of the effects of deposit insurance lies in the combination between this unique policy change and the granularity of our data. In particular, to estimate individual-level responses to the policy change, we must consider variation across time and within individuals. We achieve this by including depositor fixed effects in all our event studies and DiD specifications. The individual fixed effects also control for any time-invariant characteristics (like financial literacy, education level, or marital status). We also include month-year fixed effects, which absorb any common shocks affecting all depositors simultaneously. The combination of an exogenous change in deposit insurance and depositor fixed effects advances the empirical study of depositor behavior, which has relied on other sources of variation, like the in- and outflows of deposits following policy events

¹ Refer to the approval document of FOGAFIN's board (FOGAFIN 2017) and the newspaper article of La Republica (2017). It is important to highlight that deposit insurance matters for depositors in Colombia. As described by the International Monetary Fund (2016), deposit insurance functions in accordance with the best practices of deposit guarantee funds, repaying insured depositors within 2 days of intervention. Furthermore, this fund is actively engaged in payments. For example, FOGAFIN disbursed more than 36 billion COP (approximately US\$12 million) in 2015 following the liquidation of an insured financial institution.

(Iyer and Puri 2012; Artavanis et al. 2019) or the within-bank and across account movements using bank and account-type fixed effects (Iyer et al. 2019).

Our results confirm the predictions of the model and indicate that deposit insurance has a causal and heterogeneous effect on deposits. The following are our two main empirical findings. First, a descriptive analysis shows that depositors were bunching below the 20 million COP insurance threshold before the policy change. In particular, we observe that an extensive mass of individuals held deposits between 15 million and 20 million COP. We refer to these individuals as the “bunchers.”

Second, we test the prediction of the model that the response to changes in deposit insurance should be heterogeneous depending on the level of deposits. We start by assigning individuals to one of four bins according to their average monthly deposits before the policy change. Bin 1 includes the bunchers, that is, depositors with average deposits between 15 million and 20 million COP. Bin 2 includes individuals with 20 million to 50 million COP who are partially insured before the change and completely insured after. Individuals in bin 3 have prepolicy deposits between 50 million and 75 million COP, and those in bin 4 have deposits above 75 million. The model predicts that the response of individuals in bin 1 should be larger than that of individuals in bin 2, which in turn should be larger than that of individuals in bins 3 and 4.

Using event study and DiD designs, we quantify the differential response of individuals in each bin to the increase in deposit insurance. Our event study specification indicates that the trends of deposits across bins 1, 2, 3, and 4 are parallel before April 2017. However, in the following 6 months, deposits expand significantly more for the bunchers (bin 1) and for individuals in bin 2. Relative to individuals in bin 3, the response of individuals in bin 1 corresponds to an increase of 4.5% in the level of deposits and to 0.7 percentage points in the rate of deposit growth. Individuals in bin 2 increase their level of deposits by 3.4%. Those in bin 4 do not show any change in the level or growth rate of deposits. These results indicate that individuals below the old policy threshold respond with a higher increase in deposits. We also show that after the policy change, the share of deposits placed in savings and fixed-term deposits increases by 1.2 percentage points for depositors in bin 1.²

Our results are consistent with the mechanism highlighted by the model: with the increase in deposit insurance, depositors revise their savings decision to increase deposits since their expected return increases. But in a richer model,

² We offer a series of alternative specifications to study the robustness of these results. In particular, we show that our results are robust to focusing on the bunchers and placing all other depositors in the control group, using a continuous measure of exposure to the policy change, including branch-month-year fixed effects to account for branch time-varying shocks, and to using individuals in bin 4 as the control group instead of those in bin 3. To provide additional evidence on the lack of anticipation of the policy change, we show that the stock prices of Colombian banks evolve along parallel trends compared to other companies (insurances, utilities, etc.) before the policy change, but they increase differentially after the deposit insurance reform.

with multiple assets and returns, the change in deposit insurance would skew the asset allocation toward deposits. To explore this possibility, we present results from a survey that we conducted with the bank on a random sample of 1,000 borrowers in our data. We find that an important fraction of clients knew about FOGAFIN and the policy change and that individual asset holding changed in response to the increase in deposits with a notable decline in cash. Two alternative hypotheses could explain our results. First, our results may be driven by customers who had multiple deposit accounts across banks before the policy change, closed some of them, and then consolidated their deposits into a single one. We refer to this possibility as the “reshuffling hypothesis.” Second, it is possible that sharp changes in interest rates lead to large changes in deposits. We provide evidence that discards these two possibilities as important explanations of our findings.

Finally, we quantify the magnitude of the effect of deposit insurance on depositor behavior by estimating the elasticity of deposit growth to deposit insurance. To do so, we define a depositor-specific measure of increased insurance coverage, which is equal to the ratio between the change in insured amounts (30 million COP) and average monthly deposits before May 2017. Our estimated elasticity corresponds to 2.3%, implying that an increase in the share of insured deposits of 1 point translates into 2.3% additional deposits. To benchmark this number, we use bank-level data from the United States and the increase in the insurance threshold of 2008 that went from US\$100,000 to US\$250,000. Despite the limitations of this comparative exercise, we show that deposit growth was higher among banks with a higher share of insured deposits and find an estimated elasticity that is in line with the parameter estimated in Colombia.³

This paper contributes to the literature on deposit insurance, which received a key stimulus from the seminal work of Diamond and Dybvig (1983) showing how deposit insurance can enhance the stability of financial intermediaries. In recent years, the theoretical literature has explored the role of deposits in creating liquidity and their relation to insurance (Diamond and Rajan 2000); the relation between central banks, deposit insurance, and liquidity shortfalls (Repullo 2000); the fragility of banks to bank runs given by the nature of demand deposits (Goldstein and Pauzner 2005); the relation between deposit insurance and money market freezes (Bruche and Suarez 2010); the role of government guarantees in improving liquidity provision (Allen et al. 2018); the role of insurance and default in macro dynamic stochastic general equilibrium models (van der Kwaak, Madeira, and Palma 2021); and the conditions of an optimal deposit insurance coverage in the presence of bank runs (Davila and Goldstein 2021). Our paper contributes to this literature by formalizing, in a model in the spirit of the modern public finance literature

³ The estimated elasticity for the United States also can be valuable to policy makers who have to decide the optimal level of deposit insurance during a financial crisis.

(Chetty et al. 2011; Kleven 2016), that depositors bunch at the insurance threshold and they increase their deposits in a heterogeneous fashion when it changes.

Our paper also contributes to the empirical literature on the effect of deposit insurance on deposits. The closest paper to our work is by Martin, Puri, and Ufieri (2018), who study daily account-level data from a bank in the United States and measures the inflow and outflow of uninsured depositors after negative regulatory news. Also related to our study, Chernykh and Cole (2011) find that the introduction of deposit insurance in Russia led to an expansion in the level of retail deposits and their ratios to total assets. Iyer et al. (2019) show how changes in Danish deposit insurance in 2010 induced a reallocation of uninsured deposits away from nonsystemic to systemic financial institutions. Recently, Atmaca et al. (2020) study the case of Belgium during the global financial crisis and show how depositors react to changes in the insurance threshold depending on the ownership status of banks. Our work contributes to this literature by focusing on insured depositors and showing how they respond to deposit insurance. Crucially, we estimate the response using within-individual variation and an exogenous and unanticipated change in the insurance threshold. This allows us to estimate an effect that is unconfounded by other economic forces that are typical of periods of financial turbulence, which is novel to this literature.⁴

1. Theoretical Framework

This section presents a two-period model of depositor behavior in a simplified setting based on Davila and Goldstein (2021) with an application of the bunching framework of the public finance literature (Chetty et al. 2011; Kleven 2016).

Consider individuals with preferences defined over their consumption of a representative basket of goods at time 1 and at time 2. The utility function can be described by $U(C_1, C_2)$, where C_1 is the consumption in dollars at time 1 and C_2 is the consumption at time 2. The utility function satisfies the standard regularity assumptions, $U'(\cdot) > 0$, $U''(\cdot) < 0$, and $\lim_{c \rightarrow 0} U'(C) = \infty$.

Each individual receives an endowment $Y_i > 0$ at time 1 and decides how much to consume in this time, C_1 , and how much to deposit in the bank, D . Higher deposits lead to higher consumption at time 2, as $C_2 = \tilde{R}_D D$, with $\tilde{R}_D$ being the gross deposit rate. Individuals are heterogeneous with respect to their endowment, as captured by the density distribution $f(Y)$. Assuming that the

⁴ For other related papers on the relationship between deposit insurance and nominal deposit rates, see Gatti and Oliviero (2019) and Slutzky and Chavaz (2020). For the relation between bank runs and depositor behavior, see Iyer and Puri (2012), Iyer et al. (2016), Egan, Hortacısu, and Matvos (2017), and Artavanis et al. (2019). For papers in finance and development focusing on the role of deposits in promoting the size and depth of financial markets, see Bustos, Garber, and Ponticelli (2020), Breza and Kinnan (2021), Choudhary and Limodio (2021), and Bachas et al. (2021).

endowment distribution, preferences, and the deposit rate are smooth, then the individual optimization creates a distribution of deposits that is also smooth, as in line with Chetty et al. (2011) and Kleven (2016).

In this setting, the holding of bank deposits is risky as there is a probability $\pi \in (0, 1)$ of bank failure between time 1 and 2. In addition, the variable π is constant and exogenously determined; see Davila and Goldstein (2021) for its endogenous characterization. At the same time, the regulator offers deposit insurance on an amount $\delta_1 > 0$ of deposits. Therefore, in case of bank failure, a depositor receives the entire amount $R_D D$ if $R_D D < \delta_1$, with R_D being exogenously determined and constant. However, in case $R_D D \geq \delta_1$, the depositor receives an amount δ_1 . To keep the model simple, we restrict uninsured deposits to not being reimbursed in case of bank failure, implying a zero recovery rate. However, adding a positive recovery rate would not change the key predictions of this model.

The expected remuneration of bank deposits can be described through the following expression:

$$\tilde{R}_D D = \begin{cases} R_D D & \text{if } R_D D \leq \delta_1 \\ R_D(1 - \pi)D + \pi\delta_1 & \text{if } R_D D > \delta_1, \end{cases}$$

which states that deposits are remunerated according to R_D if the total deposits are lower than the deposit insurance threshold, $R_D D \leq \delta_1$. However, if the total deposits exceed the threshold, $R_D D > \delta_1$, then the depositor receives $R_D D$ if the bank does not fail (with probability $1 - \pi$) and δ_1 in case the bank fails (with probability π).

As a result, this structure of the deposit remuneration creates a convex kink in the individual's budget constraint at time 2 as follows:

$$C_2 = \tilde{R}_D D = R_D D - \pi \times (R_D D - \delta_1) \times I(R_D D - \delta_1), \quad (1)$$

with the consumption in time 2, C_2 , being equal to the remuneration of bank deposits for individuals below the deposit insurance threshold, $R_D D$, minus an amount lost in case of bank failure, $\pi \times (R_D D - \delta_1)$, which only involves individuals holding more than the insurance threshold, as reported by the indicator function $I(R_D D - \delta_1)$. This expression is convenient because it leads us to consider two cases in which the kink disappears: (1) a zero probability of bank failure, given by $\pi = 0$, and (2) an unlimited deposit insurance, given by $\lim_{\delta_1 \rightarrow \infty} I(R_D D - \delta_1) = 0$. The comparative static analysis in the next paragraphs compares the case with a limited δ_1 and a positive probability of bank failure, $\pi > 0$, versus a limited δ_1 and a zero probability of bank failure, $\pi = 0$.⁵

Figure 1 shows how the kinked budget constraint for individuals at time 2 affects the optimal holding of deposits through a classical example in the

⁵ This setting also could be generalized to study depositor behavior in settings with an unlimited δ_1 .

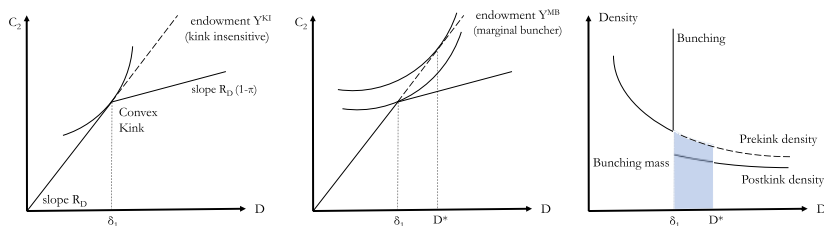


Figure 1
Kinked budget constraint and optimal deposit

This figure shows the deposit decision problem in the presence of deposit insurance for two individuals, in the left and central panels, and the distribution of deposits in the right panel. The first two panels show the kinked budget constraint: holding deposits generates a remuneration R_D when the total deposits are lower than or equal to the insurance threshold, $R_D D \leq \delta_1$ but the remuneration of deposits drops to $R_D(1-\pi)$ when the total deposits are higher than the insurance threshold, $R_D D > \delta_1$. The left panel shows the deposit decision for an individual with an endowment Y^{KI} , who is kink insensitive and would always deposit δ_1 regardless of whether the probability of bank default is zero or positive. The central panel presents the decision of an individual with endowment Y^{MB} , who is a marginal buncher. She would deposit D^* , in the presence of a zero probability of bank default, but deposits δ_1 (with $\delta_1 < D^*$) in the case of a positive probability of bank default. The right panel shows the distribution of deposits in the presence of a positive probability of bank default and limited insurance, δ_1 , labeled as Postkink density, and the distribution of deposits without such kink and zero probability of default, indicated as Prekink density. This panel shows the presence of excessive bunching at the δ_1 threshold, with the shaded area indicating the corresponding bunching mass.

bunching literature. The left and central panels present the budget set expressed by Equation (1), which generates a convex kink at δ_1 . The budget set is linear with slope R_D for $D \in [0, \delta_1]$ and $R_D(1-\pi) < R_D$ for $D > \delta_1$. The left panel shows an individual with endowment Y^{KI} and an optimal level of deposits δ_1 . This individual is labeled as “kink insensitive” as she would always deposit δ_1 both in the presence of the kink and in a counterfactual scenario in which there is a zero probability of bank default. In fact, her indifference curve is tangent both to the kinked budget set and to the counterfactual linear budget set, which is represented by a dashed line for $D > \delta_1$.

The central panel of Figure 1 shows the case of an individual with endowment Y^{MB} , who is the marginal buncher and $Y^{MB} > Y^{KI}$. In the presence of a zero probability of bank default, her deposits are determined by the tangency between the highest indifference curve and the dashed budget set, choosing the optimal deposits $D^* > \delta_1$. However, because of the kink implied by the positive probability of default, her optimal deposits are equal to δ_1 . Given the assumptions on the utility function and the smoothness of $f(Y)$, individuals with endowments in $[Y^{KI}, Y^{MB}]$ will choose an optional level of deposits of δ_1 . In other words, an excess mass of depositors choose $D^* = \delta_1$ and therefore “bunch” at δ_1 .

The right panel of Figure 1 depicts the distribution of deposits. The solid line represents the case with $\pi > 0$ and the dashed line the case with $\pi = 0$ (for $D > \delta_1$), labeled “Prekink density.” Because of the kink induced by δ_1 , a mass of individuals bunch at the δ_1 threshold instead of choosing an optimal amount of deposits between δ_1 and D^* . This bunching mass is indicated through the shaded area of the panel. The distribution of deposits in the presence of the kink

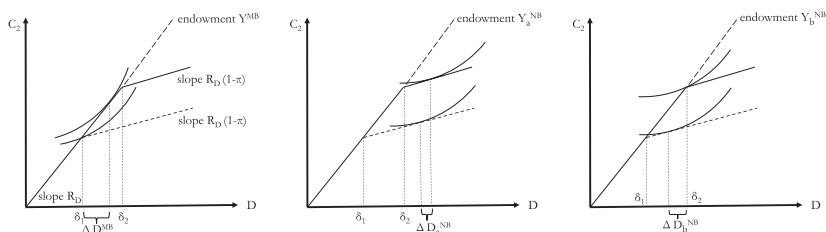


Figure 2

Depositor response to an increase in deposit insurance

This figure shows the deposit decision problem in the presence of a positive probability of default and when the threshold of deposit insurance increases from δ_1 to δ_2 for three individuals. The panels show the change in the kinked budget constraint. After the increase in insurance, depositors with initial optimal deposits $D \in (\delta_1, \delta_2)$ experience an increase in the deposit rate from $R_D(1-\pi)$ to R_D . The left panel shows the deposit decision for the marginal buncher, with endowment Y^{MB} , who deposits an additional amount ΔD^{MB} in response to the increase in insurance. The center panel shows the deposit decision for a nonbuncher, with endowment Y_a^{NB} , who deposits an additional amount ΔD_a^{NB} in response to the increase in insurance. The right panel shows the deposit decision for a nonbuncher, with endowment Y_b^{NB} , who deposits an additional amount ΔD_b^{NB} in response to the increase in insurance. In the figure, $\Delta D^{MB} > \Delta D_b^{NB} > \Delta D_a^{NB}$.

(labeled as “Postkink density”) also exhibits a discontinuity in the distribution at δ_1 . These findings are summarized by the following proposition:

Proposition 1. In the presence of a positive probability of bank failure, $\pi > 0$, a limited threshold of deposit insurance creates a kink in the expected gross deposit rate and induces bunching in the deposit distribution at the threshold.

1.1 Deposit behavior and an increase in deposit insurance

This conceptual framework is useful to study how depositors respond to an increase in deposit insurance, which is the policy under consideration in this paper. The expression of the expected gross deposit rate, $\tilde{R}_D D = R_D D - \pi \times (R_D D - \delta_1) \times I(R_D D - \delta_1)$, implies that an increase in deposit insurance from δ_1 to δ_2 increases the deposit rate and promotes deposits for depositors with amounts higher than or equal to the old threshold. At the same type, our paper highlights a key heterogeneity: individuals bunching at the old threshold exhibit the strongest increase in deposits, and individuals with deposits between δ_1 and δ_2 before the policy change increase their deposits more than those holding amounts higher than δ_2 .

Internet Appendix B characterizes these statements analytically through an extension of the model that assumes a constant relative-risk aversion utility function. Here, we discuss the results graphically with the help of Figure 2. The left panel shows how the marginal buncher, with endowment Y^{MB} (presented in the central panel of Figure 1), would react to an increase in deposit insurance. As the threshold increases from δ_1 to δ_2 , the individual’s budget set expands since for $D \in (\delta_1, \delta_2)$ the remuneration is now R_D (rather than $R_D(1-\pi)$), and the new kink moves to the right of the old kink. The new level of deposits

is given by the tangency between the highest indifference curve and the expanded budget set, leading to an increase in deposits denoted by ΔD^{MB} . The central panel of Figure 2 presents the case of an individual with a higher endowment Y_a^{NB} , who was not bunching at the old threshold and with initial optimal deposits above δ_2 . Her optimal level of deposits increases by ΔD_a^{NB} . Finally, the right panel presents the case of an individual with endowment Y_b^{NB} with initial optimal deposits in (δ_1, δ_2) . In this case, the optimal level of deposits increases by ΔD_b^{NB} . Figure 2 shows that the deposit expansion for individuals who were marginal bunchers is stronger than that of individuals with initial deposits in (δ_1, δ_2) , as $\Delta D^{MB} > \Delta D_b^{NB}$. Furthermore, the response of the individual with deposits in (δ_1, δ_2) is larger than that of the individual with initial deposits above δ_2 (as $\Delta D_b^{MB} > \Delta D_a^{NB}$). The following proposition summarizes these findings:

Proposition 2. An increase in the deposit insurance threshold leads to a rise in deposits for individuals with deposits higher than or equal to the old threshold. Individuals who were bunching at the old threshold exhibit the largest increase in deposits, and individuals between the old and the new threshold expand their deposits more than individuals with deposits above the new deposit threshold.

This theoretical section offers a stylized model that guides our empirical analysis and is meant to capture the effect of deposit insurance on depositors in partial equilibrium and in the short run. We acknowledge, though, that a general equilibrium framework would endogenize the effect of deposit insurance on the probability of bank default. This consideration goes beyond the scope of this paper and we refer the reader to Davila and Goldstein (2021) for a framework that studies deposit insurance in a general equilibrium setting. It is also important to note that our model is built around an intertemporal endowment allocation decision, in which an individual makes a savings choice that is mapped only into a decision on deposits. A richer version of the model would feature the choice of multiple assets, each with a risk-adjusted return. In that case, a higher insurance threshold would also increase the risk-adjusted return of deposits but the corresponding increase in deposits would emerge from a different composition of savings, in line with the findings reported below in Section 4.1. We use a simple setting to guide our empirical analysis and leave a more complex multiasset model for further investigation. We now turn to testing the predictions of the model and to estimating the causal effect of deposit insurance on depositor behavior.

2. Setting and Data

This section describes the context, the data of our study, and how we identify bunchers. The context is special in that it presented a change in the deposit

insurance threshold that was large and unrelated to economic conditions. Furthermore, our data are unique in that their frequency is high and the level of cross-sectional variation is the depositor.

2.1 FOGAFIN and deposit insurance

Our study takes place in Colombia, a middle-income country with a track record of prudent macroeconomic and fiscal management. Like other countries in Latin America, Colombia is subject to commodities and export fluctuations, which in turn can lead to downturns (Reinhart and Wickham 1994). However, this country maintained its investment-grade rating from 2013 to 2021 and a record of robust economic growth.

Colombia has a large and diversified financial system compared with peer countries, with significant cross-border institutions and large conglomerates. The four largest banking groups are considered systemic and account for over 70% of banking assets. Of these four groups, the top three are domestic and privately owned. The assets under management by financial companies are close to 150% of gross domestic product (GDP), with banks accounting for about 45% of the total.⁶ As discussed by International Monetary Fund (2016), banks are strongly funded with local deposits.

In this setting, the Colombian deposit guarantee fund (FOGAFIN) is the financial authority responsible for protecting the accounts of the public in commercial banks, all of which are required to register with it. This financial authority administers the deposit insurance scheme; namely, it guarantees that depositors will recover all or part of their money (up to 50 million COP per individual and per institution) and pays out to account holders if an institution is liquidated.⁷ FOGAFIN was created in 1985 to deal with a crisis, during which it intervened in the process of capitalizing the financial system (Perez-Reyna and Osorio 2018). Over the last 30 years, it has gained a central role in handling the difficulties faced by the financial system and has taken part in the winding up of financial institutions, paying out deposit insurance, and following up on the liquidation of financial entities. In 2015, it intervened and spent 36 billion COP (approximately US\$12 million) to repay insured deposits of *Internacional*, a Colombian financing company. Overall, FOGAFIN's role is analogous to the Federal Deposit Insurance Corporation (FDIC) in the United States, and it functions according to the best practices of deposit guarantee funds, being able

⁶ Of 57 credit institutions, 25 are banks (the others are financial corporations of various types as well as financial cooperatives).

⁷ For individual accounts, FOGAFIN pays the insured amount to the account holder and the spouse is not entitled to the insurance. The two types of joint accounts in Colombia are: "collective accounts" and "joint accounts." Collective accounts can be used by any account holder without the signature of the other account holders. For joint accounts, the signature of all account holders is required to use the product. For collective accounts, the insured sum is given to the first person to apply for reimbursement. For joint accounts and in the event of an application for reimbursements, the total amount at the time of bank default is divided by the number of account holders. Then each account holder is reimbursed the difference between 50 million COP and this number.

to pay insured depositors within two days from the intervention of a financial institution.

The reimbursement process is easy and accessible. It involves four steps, two of which must be done by depositors: completing an online form—or a physical form for individuals without internet access—after the previous contact from FOGAFIN if eligibility requirements are met and presenting the form to the institution. Last, the institution arranges the payment of the insured amount through an electronic funds transfer, a check, or cash in two to four days.⁸

Moreover, deposit insurance in Colombia only covers savings products, meaning it excludes any fiduciary contracts, bonds, stocks, and insurance products. It is mandatory for every commercial bank in Colombia to be registered with FOGAFIN, so every savings account in every commercial bank and financial corporation, public or private, is protected. Finally, it is important to clarify that the deposit insurance scheme covers all deposits up to the threshold per person and per financial institution. If an individual owns two deposit accounts, each in a different financial institution, each account is covered up to 50 million COP.⁹

2.2 The policy change

Central to our study is the unexpected policy change, which occurred in 2017 and involves the maximum amount insured by FOGAFIN. On April 18, 2017, the Colombian deposit guarantee fund increased the insurance for individual deposits from 20 million COP (approximately US\$6,780 at the average exchange rate of around 2,950 COP per US\$) per person, per institution, to 50 million COP (approximately US\$16,950). Importantly, this policy change was unexpected, and the announcement was made to the public only one day after the implementation.

This increase aimed to update the real value of the insurance threshold that had not changed since the year 2000, contrary to most policy changes that usually happen during periods of financial turmoil. According to the Financial Superintendent of FOGAFIN, Jorge Castaño, “the 20 million COP of the year 2000 corresponds to the 50 million COP of today.”¹⁰ In sum, the policy change induced important variation in the deposit insurance threshold (it more than doubled overnight) that was exogenous to the health of Colombia’s banking system. Moreover, individual depositors were automatically granted the increment without any action required on their part. These specific features

⁸ FOGAFIN charges all associated financial institutions a premium of 0.3% per year of the liabilities covered. The accounts that are covered by deposit insurance, and thus affected by the policy change, are all the savings products of financial institutions registered with FOGAFIN, whether owned by individuals or firms. For more information, see FOGAFIN (2022).

⁹ For additional details on FOGAFIN and the deposit insurance scheme, such as the reimbursement process or the list of insured products, see FOGAFIN (2022).

¹⁰ See La Republica (2018), Revista Semana (2017), and Presidencia de la República (2017).

of the policy change offer a unique opportunity to investigate the effect of deposit insurance on depositors and banks.

2.3 Data

We are interested in the effect of the exogenous increase in deposit insurance on depositor behavior. Our model predicts that an increase in deposit insurance leads individuals to increase their savings.¹¹ Therefore, testing the predictions of the model requires panel data at the borrower level that allows studying variation over time and within individuals. The data from the bank that we describe in this section are ideal for this task.

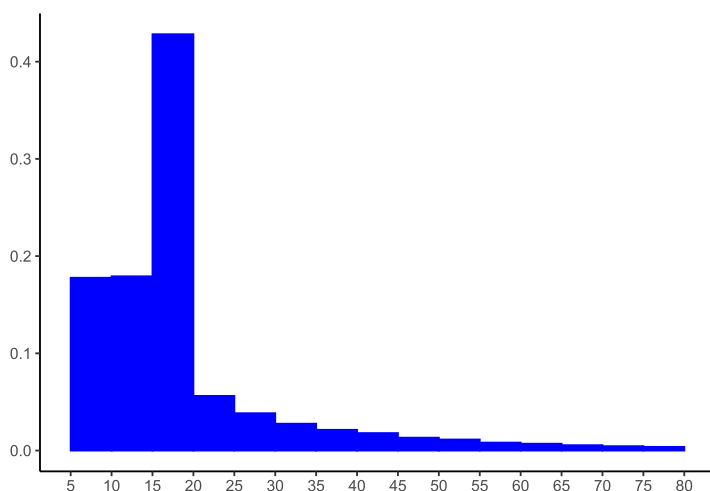
We use individual-level data on monthly deposits from a large Colombian bank. We observe all accounts of individuals in the bank and focus our analysis on depositors with two characteristics. First, they had an active account in the 24-month window around April 2017, hence from April 2016 to March 2018. Second, they had a monthly balance of at least 5 million COP (around US\$1,700) in the period of analysis. We impose this second restriction to screen out from the exercise accounts dedicated to microfinance and micro-savings. We then trim individuals with an average level of deposits in the period April 2016 to April 2017 in the top 1% to reduce the possibility of spurious outliers, as is standard in the literature. This results in a sample of 59,052 individuals.

2.4 Identifying bunchers

We now turn to the identification of bunchers in the data. Our model predicts that the threshold of deposit insurance leads a mass of depositors to hold the maximum amount of deposits insured. Before the policy change, this threshold was 20 million COP. Figure 3 presents the distribution of monthly average deposits from April 2016 to April 2017 and shows evidence of bunching below the 20 million threshold, specifically at the 15 million to 20 million bin. The presence of bunching in the distribution of bank deposits points to the role of the deposit insurance threshold in shaping depositor behavior. We will refer to the individuals in this bin as the “bunchers.” While this is only a descriptive result, it is consistent with depositors being aware of the discontinuity in deposit insurance and allocating their portfolios optimally. Furthermore, it is consistent with Prediction 1 of our model.

Another prediction of our model is that the response of deposits to a change in deposit insurance is heterogeneous and depends on the level of deposits before the policy change, relative to the old and new thresholds. To test this prediction empirically and estimate the heterogeneous response of deposits to changes in deposit insurance, we define the following bins: *bin 1*: individuals with average monthly deposits from April 2016 to April 2017

¹¹ In the model, this increase in savings comes at the expense of present consumption. But in reality, the increase is more likely to come from the liquidation of other assets. We will discuss this issue further in Section 4.

**Figure 3****Distribution of average deposits and bunching**

This figure presents the distribution across individuals of average deposits in millions of COP in the period before the policy change (April 2016–April 2017). The threshold of insured deposits before the change was 20 million COP.

between 15 million and 20 million COP, the bunchers; *bin 2*: individuals with average monthly deposits before the policy change between 20 million and 50 million COP; *bin 3*: individuals holding between 50 million and 75 million COP; and *bin 4*: individuals holding 75 million COP or more.

Proposition 2 predicts that in response to an increase in the deposit insurance threshold, the bunchers (*bin 1*) will increase their deposits to a larger extent than depositors in *bin 2*. In turn, depositors in *bin 2* will generate larger deposit inflows than individuals in bins 3 or 4. We now turn to testing these predictions in event study and DiD designs. We will use the change in deposit insurance from 20 million to 50 million COP as an exogenous source of variation to identify the differential response of depositors in each bin.

3. Effect of Deposit Insurance on Depositor Behavior

This section presents our empirical analysis that exploits the change in the insurance threshold in April 2017 to study the effect of deposit insurance on depositor behavior. We are interested in the effect of changes in the deposit insurance threshold on the individual responses that we capture with three variables: the natural logarithm of the monthly deposited amount (*log deposits*), the monthly growth rate of deposits (*Deposit growth*, defined as the difference in log deposits between month t and month $t - 1$), and the share of deposits in savings and fixed-term accounts (*Savings share*). Table 1 shows summary statistics of these variables.

Table 1
Descriptive statistics

	Mean (1)	SD (2)	25th pct. (3)	Median (4)	75th pct. (5)
log(Deposits)	16.88	0.53	16.56	16.60	17.11
Savings share	0.87	0.30	1	1	1
Deposit growth	0.000	0.13	0	0	0.003

This table reports descriptive statistics of the main variables in our analysis. To construct the sample, we start with a monthly panel at the depositor level and consider individuals who had an active account with at least 5 million COP every month from April 2016 to March 2018. We remove from the panel individuals in the top 1% according to their average level of deposits from April 2016 to April 2017. We then discard individuals with less than 15 million COP and assign the remaining individuals to bins according to their average monthly deposits from April 2016 to April 2017: between 15 million and 20 million COP (bin 1, the “bunchers”), between 20 million and 50 million COP (bin 2), between 50 million and 75 million COP (bin 3), and above 75 million COP (bin 4). There are 24,727 individuals in bin 1, 10,249 in bin 2, 2,155 in bin 3, and 1,408 in bin 4. *log(Deposits)* is the natural logarithm of individual total monthly deposits; *Deposit growth* is the logarithmic difference of total monthly deposits between t and $t - 1$; and *Savings share* is the share of savings and fixed-term deposits over total deposits.

To study the effect of change in deposit insurance, we start with an event study design analyzing 12 months around the policy change followed by a DiD specification to estimate the average effect. We then present a battery of robustness exercises.

3.1 Event study

The first specification is an event study design around April 2017, the month of the change in the deposit insurance threshold. This allows us to investigate whether depositors belonging to different bins were on parallel trends before the policy and to study its dynamic effects. Importantly, the specification includes depositor fixed effects so that our coefficients are identified from within-individual variation and can therefore be interpreted as capturing the behavioral changes of individuals.

We estimate by ordinary least squares (OLS) the following specification using individuals in bins 1–4 and in a sample period that starts in November 2016 and ends in November 2017, to consider 6 months before and after the policy change:

$$\begin{aligned} y_{ibt} = & \sum_{\tau=-6, \tau \neq -1}^{\tau=6} \alpha_{1\tau}(X_{\tau} \cdot Bin_1) + \sum_{\tau=-6, \tau \neq -1}^{\tau=6} \alpha_{2\tau}(X_{\tau} \cdot Bin_2) \\ & + \sum_{\tau=-6, \tau \neq -1}^{\tau=6} \alpha_{4\tau}(X_{\tau} \cdot Bin_4) + \iota_i + \mu_t + e_{ibt}. \end{aligned} \tag{2}$$

y_{ibt} is the log of deposits of individual i , in bin b at the end of the month-year period t . The term Bin_b is equal to 1 if individual i is in bin b and 0 otherwise, based on its average amount of prepolicy deposits. X_{τ} is a lead and

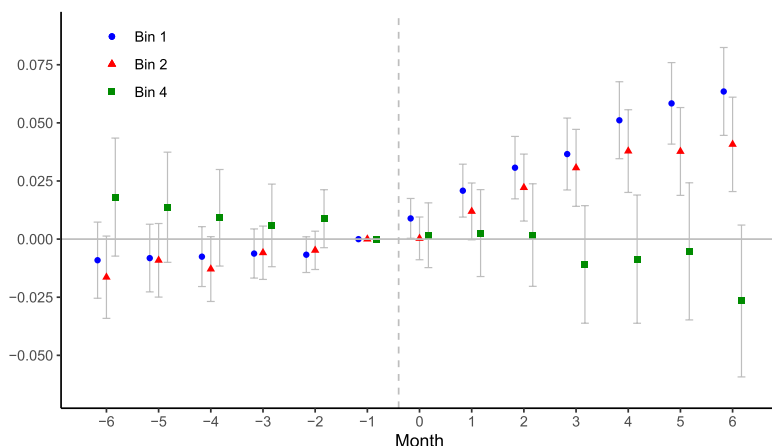


Figure 4
Effect of deposit insurance, event study

This figure reports results from the event study of Equation (2). The y-axis reports the coefficients of the effect of the policy change on the log of deposits, for each bin. The x-axis reports the corresponding lead or lag, where zero corresponds to May 2017. See the notes of Table 1 for the description of the sample and the bins. Vertical lines represent 95% confidence intervals. Individuals in bin 1 had average monthly deposits from April 2016 to April 2017 between 15 million and 20 million COP, individuals in bin 2 between 20 million and 50 million COP, in bin 3 between 50 million and 75 million COP, and in bin 4 above 75 million COP.

lag indicator.¹² Note that we omit Bin_3 from the equation and X_{-1} from each summation. ι_i and μ_t denote fixed effects, respectively, for each individual and month of each year, henceforth month-year. e_{ibt} denotes the error term, which we cluster at the individual level. $\alpha_{b\tau}$ is the coefficient of interest, and it expresses the change of deposits in a specific bin during a given month-year t , relative to deposits in the same month-year in bin 3. Therefore, identification of the $\alpha_{b\tau}$ coefficient comes from the differential change in deposits relative to the omitted bin 3.

Figure 4 shows the evolution of the $\alpha_{b\tau}$ coefficients across bins 1, 2, and 4 before and after the policy change was implemented. The figure reveals two important elements. First, before the policy change, the evolution of deposits across bins is on parallel trends. For all bins in the preperiod, we cannot reject the null hypothesis that deposits of a specific bin are on the same trend as depositors in bin 3. Second, individuals belonging to bin 1 substantially increase their deposits after the policy change relative to depositors in bin 3. This increment is steady and persistent. Depositors in bin 2 react less starkly but still increase their deposits. Relative to depositors in bin 3, depositors

¹² X_0 equals one for the first month after the start of the policy change, that is, for May 2017. X_1 to X_6 equal one for June 2017 to November 2017, respectively. Similarly, X_{-1} equals one for April 2017, and X_{-2} to X_{-6} equal one for March 2017 to November 2016.

in bin 4 do not behave statistically differently.¹³ These results are consistent with the predictions in Proposition 2 and show that the response to deposit insurance is higher for depositors who are more affected by the policy change (the bunchers).

3.2 Difference-in-differences

In this section, we study the magnitude of the effect of the policy change by considering a DiD design using the same time period of the event-study specification. Compared to the event study, it assumes a constant treatment effect over time and provides a measure of the average causal effect of the increase in deposit insurance. This exercise assumes that in the absence of the treatment, depositors in the different bins would have followed parallel trends. This is in line with the findings from our event study exercise.

We estimate by OLS the following model:

$$y_{ibt} = \sum_{b=1, b \neq 3}^4 \gamma_b \text{Bin}_b \times \text{Policy}_t + \iota_i + \mu_t + \varepsilon_{ibt}, \quad (3)$$

where i denotes individuals, b denotes deposit bins, t denotes month-year periods, and y_{ibt} denotes the outcome of interest. We consider three outcomes: log deposits, monthly deposit growth (defined as the difference in log deposits between t and $t - 1$), and the share of deposits in savings or fixed-term deposits relative to total deposits.¹⁴ Policy_t is a dummy equal to zero before May 2017 and equal to one afterward.¹⁵ ι_i and μ_t denote individual and time fixed effects, respectively. ε_{ibt} denotes the error term, which we cluster at the individual level. The effect of the policy change is identified from the differential change of the outcome across deposit bins. As in the event study of the previous section, the sample consists of individuals in bins 1, 2, 3, and 4 and the months from November 2016 to November 2017.

Table 2 presents the results of estimating Equation (3). Column 1 shows that deposits in bins 1 and 2 increase after the policy (relative to deposits of bin 3) by 4.5% and 3.4%, respectively. This increase is large and significant at the 1% confidence level. Depositors in bin 4 do not exhibit a differential change in deposits relative to depositors in bin 3. Column 2 shows that the growth rates

¹³ The point estimates of bin 4 show a slight pre-trend which may reflect the fact that individuals in bin 4 belong to some of the wealthiest groups, whose income and wealth shocks may deviate from the rest of the individuals and in particular from those in bin 3. However, none of these coefficients is statistically different from zero and we cannot reject the hypothesis of parallel trends.

¹⁴ Our model does not yield any predictions on the savings share of deposits. Nevertheless, if we extend the model to a multiperiod problem, where individuals can choose between deposit contracts with different maturities, we expect an increase in deposit insurance to increase the maturities chosen.

¹⁵ Although the policy was implemented on April 18, 2017, we code $\text{Policy}_t = 1$ for t in May 2017 and onward because this is the first month during which the policy was in place every day. Our results are robust to coding $\text{Policy}_t = 1$ also for April 2017.

Table 2
The effect of deposit insurance, DiD

	log(Deposits) (1)	Deposit growth (2)	Savings share (3)
$Bin_1 \times Policy_t$	0.045*** (0.008)	0.007*** (0.002)	0.012*** (0.005)
$Bin_2 \times Policy_t$	0.034*** (0.009)	0.002 (0.002)	0.0005 (0.005)
$Bin_4 \times Policy_t$	-0.016 (0.011)	-0.002 (0.004)	0.006 (0.007)
Depositor FE	Yes	Yes	Yes
Month FE	Yes	Yes	Yes
Obs.	501,007	501,007	501,007
Adj. R^2	.906	-0.035	.853
p -value	.000	.000	.000

This table reports estimates from the difference-in-differences specification of Equation (3). Each column corresponds to a different regression using the outcome listed in the column header. All regressions include depositor and month-year fixed effects. The reported coefficient corresponds to the interaction between a bin dummy and the policy dummy, $Policy_t$, equal to one from May 2017 onward. See the notes of Table 1 for the description of the sample and the dependent variables. Standard errors clustered at the depositor level are reported in parentheses. The p -value reported in the last row corresponds to a test with the null hypothesis that the coefficient of $Bin_1 \times Policy_t$ is equal to the coefficient of $Bin_2 \times Policy_t$. Individuals in bin 1 had average monthly deposits from April 2016 to April 2017 between 15 million and 20 million COP, individuals in bin 2 between 20 million and 50 million COP, in bin 3 between 50 million and 75 million COP, and in bin 4 above 75 million COP. $*p < .10$; $**p < .05$; $***p < .01$.

of bin 1 increased after the policy change by 0.7 percentage points. We find no effect in deposit growth for bins 2 or 4. Column 3 shows that depositors in bin 1 increase their holding of deposits in savings and fixed-term accounts by 1.2 percentage points. However, the saving shares of depositors in the other bins are unaffected. For the three variables in the table, the difference between the coefficients of bin 1 and bin 2 is statistically significant and important in magnitude, especially for deposit growth and the savings share.¹⁶ This is consistent with Proposition 2 of the model which states that, in response to an increase in deposit insurance, bunchers increase their deposits more than individuals just above the threshold.

Overall, these results indicate that the increase in the deposit insurance threshold in Colombia led to significant changes in depositors' behavior, and, consistent with Proposition 2, this change in behavior is heterogeneous across bins.

3.3 Robustness

In this section, we investigate the robustness of our results on the effect of deposit insurance on depositor behavior. We first estimate the effect of the policy if we use as the control group the individuals in bins 2, 3, and 4 taken together. This control group is problematic since it includes individuals in bin 2,

¹⁶ Table 2 reports the p -value of a test of the null hypothesis that the coefficient of bin 1 is equal to the coefficient of bin 2. For the three variables in the table, we reject the null at a significance level smaller than 0.1%.

who are also importantly affected by the policy. Still, it allows us to ask if the response of the bunchers is indeed larger than the response of individuals in other bins, as predicted by our model. Internet Appendix Figure A1 presents the results from the corresponding event study and shows that the bunchers were on parallel trends compared to individuals in bins 2 to 4. After the policy change, the bunchers increase their deposit holdings compared to individuals in this control group. As shown in Internet Appendix Table A1, the average effect on total deposits using individuals in bins 2 to 4 as controls is 2.1%.

In a second exercise, we account for shocks at the branch level that can potentially bias our estimates of the effect of deposit insurance. We do so by including a set of branch-month-year fixed effects. Our panel contains 744 branches distributed across Colombia. This set of fixed effects accommodates time trends and shocks that are specific to the branch. Internet Appendix Table A2 reports the results, which are very similar to the ones reported in the baseline exercises (Table 2). In this case, the effect of the policy on depositors in bin 1 is larger for the log deposits variable but is in line with the previous estimates for deposit growth.¹⁷ The effect on the savings share remains positive but is not statistically significant.

In the third exercise, we study the effect of deposit insurance using a continuous measure of exposure to the policy change. For each individual, we define the $\Delta Coverage_i$ variable as the ratio between the change in available insurance, 30 million COP, over the average level of deposits in the prepolicy period. Individuals with lower average deposits before the policy see a larger increase in the fraction of insured deposits, for the same change of 30 million COP, and therefore have a higher $\Delta Coverage_i$. Internet Appendix Table A3 reports the results of estimating Equation (3) but replacing the $\gamma_b Bin_b \times Policy_i$ terms by the single term $\gamma \Delta Coverage_i \times Policy_i$. The table shows that individuals more exposed to the policy increase more their deposits (column 1), have a higher rate of deposit growth (column 2), and have a higher savings share (column 3). These findings show that our baseline estimates of the effect of deposit insurance are robust to using a continuous measure of exposure to the policy change.

In a fourth exercise, we see if our results depend on the choice of the bin for the control group. Since individuals in bin 4 are also above the threshold after the policy change, they also constitute a plausible control group. Internet Appendix Table A4 reports the results of estimating Equation (3) using individuals in bin 4 as control. Relative to bin 4, the bunchers (individuals in bin 1) increase their deposits by 6.1% and individuals in bin 2 by 5%. As expected, there is no difference in deposits if we compare individuals in bins 3 and 4.

¹⁷ This result is consistent with the occurrence of branch-level shocks that are negatively correlated with deposit behavior in bin 1, leading to a downward bias in the estimate of the γ_1 coefficient. For example, investment opportunities could change differentially across areas of the country. Therefore, our main results should be interpreted as a lower bound for the effect of deposit insurance on the behavior of individuals in bin 1.

3.4 Lack of anticipation

The results of the event study reported in Figure 4 show that depositors of the different bins were in parallel trends, implying that the policy was not anticipated. Had it been anticipated, the deposits of individuals in different bins should have started to differ before the policy change. The lack of anticipation is an important assumption that we need to estimate the policy's average treatment effect in the DiD specification. In this section, we present an additional result that supports this assumption.

The exercise is based on the idea that if the change was indeed unexpected, the prices of banks should not differ from those of other types of companies before the policy change.¹⁸ To study the effect of the policy change on stock prices we compile a data set of daily stock prices of 19 Colombian companies operating in different sectors (financial, energy, communications, and industrial, among others) from Bloomberg.¹⁹ Among these, seven are banks.

We estimate an event study specification where the outcome is the price of the stocks:

$$p_{idt} = \sum_{t=-3, t \neq 0}^3 \gamma_t (X_t \times Bank_i) + \iota_i + \mu_t + \varepsilon_{it}, \quad (4)$$

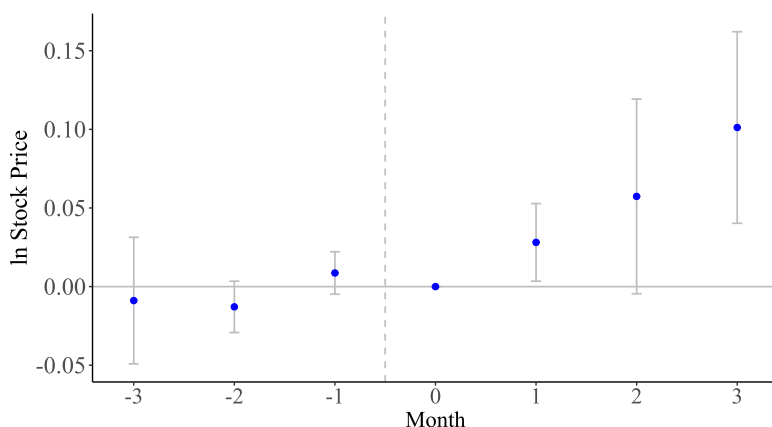
where p_{idt} is the stock price of company i in day d of month t . X_t is a lead and lag month indicator. We consider $t = -3, -2, -1, 0, 1, 2, 3$. We code $X_0 = 1$ for April 19, 2017, the first day the policy was in place; X_{-1} for the days between March 19 and April 18; and $X_1 = 1$ for the days between April 20 and May 19. We code X_2, X_{-2}, X_3 , and X_{-3} analogously and omit $t = 0$, which corresponds to April 18, 2017, the day of the policy announcement. $Bank_i$ is an indicator variable for banks. Company and month fixed effects are denoted by ι_i and μ_t , respectively. Standard errors, which we denote by ε_{it} , are clustered at the company level.

Figure 5 plots the estimates of γ_t . The figure shows that the stock prices of banks and other types of companies were on parallel trends before the policy. After April 2017, a relative increase in the stock price of banks reaches 10% three months after the policy change. This result is consistent with the policy inducing a subsidy to financial institutions that creates value by unlocking additional funding.²⁰ While the extent of this increase may be interpreted in

¹⁸ We don't have a prior on the effect of the policy change on the price of banks as its relation to bank balance sheets and profits goes beyond our theoretical framework and the scope of this paper.

¹⁹ This figure is due to the overall number of companies with available daily stock prices in Bloomberg.

²⁰ In heterogeneity exercises not reported in the paper, we find that this effect does not depend on bank fundamentals like assets or the return on assets.

**Figure 5****Deposit insurance and stock prices**

This figure reports results from the event study of Equation (4) of the effect of the policy change on the daily log price of the stock of Colombian companies. The sample contains the daily prices of 19 companies that had daily prices listed on Bloomberg. Each reported coefficient corresponds to the interaction of $Financial_t$, an indicator equal to one for financial companies, and X_t , a lead and lag indicator. We code $X_0=1$ for April 19, 2017, X_{-1} for the days between March 19 and April 18, and $X_1=1$ for the days between April 20 and May 19. We code X_2 , X_{-2} , X_3 , and X_{-3} analogously. Vertical lines represent 95% confidence intervals.

various manners, the results reported in the figure constitute evidence of the lack of anticipation of the policy change.²¹

3.5 Borrower characteristics and borrower response

In the empirical analysis presented thus far, we have followed the model's prediction to identify those individuals who respond more to the policy change. According to our stylized model, individuals responding more to the change in the deposit insurance threshold are the bunchers, that is, individuals in bin 1, who hold between 15 million and 20 million COP. We identified the bunchers based on the distribution of average deposits in the period before the policy change. But this raises the following concern: are the bunchers the only individuals who care about deposit insurance and who respond to the policy change? It might be possible to identify borrowers who respond to deposit insurance with an alternative strategy that is less reliant on our model.

In this section, we explore this possibility by investigating if the effect of the policy change on log deposits differs according to different borrower characteristics, some of which could be correlated with borrower sophistication. More sophisticated individuals should be more informed or care more about the change in the deposit insurance threshold. We consider different characteristics

²¹ In Internet Appendix C, we perform a cumulative abnormal returns (CAR) analysis with the same stock price data that we use in this section. The analysis shows that there were no differences in the CAR of banks and of nonfinancial companies before the policy change. After the policy change, the CAR of banks increases relative to that of nonfinancial companies. This result again supports the assumption of lack of anticipation.

that according to the previous literature could be correlated with borrower sophistication. First, and following Iyer and Puri (2012), we define a dummy equal to one for depositors with loan products from the bank. Second, we define a variable that captures the seniority of the bank client: the number of years the borrower has had a savings product with the bank. Third, we define a dummy equal to one if the borrower had more than one savings product type during the sample period.²² We then investigate the heterogeneous effects of the policy change along these dimensions.²³

Columns 1–3 of Table 3 report the results. Each column corresponds to a single regression with the coefficient of each bin interacted with the corresponding borrower characteristic. Two results stand out. First, and similar to the results above, the policy change leads to an increase in deposits of bins 1 and 2 relative to bin 3. Second, there are no heterogeneous effects in any of the bins across any of the heterogeneity dimensions considered. It is not the case that among individuals in bin 1, those with a loan product, higher seniority, or with more types of savings products respond more to the policy change. In other words, once we condition on a borrower being in bin 1, there is no differential response in log deposits along different dimensions that could be correlated with borrower sophistication. The same is true if we condition on any other bin. This result suggests, in line with the model, that bunchers are indeed those individuals who respond more to the policy change.

The absence of heterogeneous responses across the individual characteristics considered in this section is also important for external validity. Since the differential response across bins does not appear to be driven by specific groups within these bins, it is more likely that our results apply to customers in other banks and in other countries.

4. Mechanisms and Alternative Explanations

In this section, we discuss the potential mechanisms behind the increase in deposits. Our model postulates that a positive probability of bank default and an increase in the deposit insurance threshold lead individuals to increase their deposits. In the model, the mechanism that drives this increase is that individuals sacrifice consumption in period 1 since the expected returns of deposits are higher. But, in reality, other sources are more likely to drive the increase in deposits. We study them in this section. First, we discuss evidence from a survey that suggests that our results could be in part driven by borrowers liquidating other assets to increase their deposits. Second, we

²² There are three product types in our sample: checking accounts, savings accounts, and fixed-term deposits.

²³ Internet Appendix Table A6 presents descriptive statistics of these variables across individuals for the whole sample and for each separate bin. Importantly, the table shows that within bins there is considerable variation in each of the heterogeneity variables. Furthermore, as we move from bin 1 to bin 4, individuals are more likely to have a loan product with the bank, have higher seniority, and are more likely to have multiple savings products with the bank.

Table 3
The effect of deposit insurance, DiD, heterogeneity

	Borrower characteristics			Competition	
	Has loan (1)	Seniority (2)	Multiple products (3)	# banks (4)	75th pct. (5)
$Bin_1 \times Policy_t$	0.042*** (0.008)	0.056*** (0.012)	0.053*** (0.011)	0.062*** (0.010)	0.060*** (0.010)
$Bin_2 \times Policy_t$	0.037*** (0.008)	0.048*** (0.013)	0.042*** (0.012)	0.032*** (0.009)	0.032*** (0.010)
$Bin_4 \times Policy_t$	-0.010 (0.012)	-0.007 (0.020)	-0.022 (0.019)	-0.013 (0.013)	-0.009 (0.014)
$Bin_1 \times Policy_t \times H_i$	0.016 (0.020)	-0.0003 (0.001)	-0.002 (0.016)	-0.0001 (0.001)	-0.010 (0.016)
$Bin_2 \times Policy_t \times H_i$	-0.014 (0.021)	-0.001 (0.001)	-0.016 (0.015)	0.001 (0.001)	0.004 (0.016)
$Bin_4 \times Policy_t \times H_i$	-0.034 (0.032)	-0.001 (0.001)	0.007 (0.023)	-0.001 (0.002)	-0.019 (0.024)
$Policy_t \times H_i$	-0.016 (0.020)	0.002** (0.001)	0.019 (0.014)	-0.001 (0.001)	-0.012 (0.015)
Depositor FE	Yes	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes	Yes
Obs.	501.007	501.007	501.007	501.007	501.007
Adj. R^2	.907	.907	.907	.907	.907

This table reports estimates from the difference-in-differences specification presented in Equation (3), allowing for the effect of the policy change to depend on the heterogeneity variable H_i . The outcome is log deposits. Each column corresponds to a single regression using the heterogeneity variable listed in the column header. In column 1, H_i is a dummy equal to one for depositors who have a loan product in the sample period. In column 2, H_i is the number of years since the borrower had a savings product with the bank. In column 3, H_i is a dummy equal to one if the borrower had more than one savings product type during the 12 months prior to the policy change. The three product types are: checking accounts, savings accounts, and fixed-term deposits. In column 4, H_i is the number of competing banks operating in the municipality of the borrower's bank branch. To individuals with deposit accounts in different branches, we assign the branch with the highest level of deposits in April 2016. In column 5, H_i is a dummy equal to one if the number of competing banks is above the 75th percentile of the number of competing banks across branches. All regressions include depositor and month-year fixed effects. See the notes of Table 1 for the description of the sample and the dependent variables. Individuals in bin 1 had average monthly deposits from April 2016 to April 2017 between 15 million and 20 million COP, individuals in bin 2 between 20 million and 50 million COP, in bin 3 between 50 million and 75 million COP, and in bin 4 above 75 million COP. Standard errors clustered at the depositor level are reported in parentheses. * $p < .10$; ** $p < .05$; *** $p < .01$.

discuss an alternative explanation that highlights deposit reshuffling across banks. Third, we consider the possibility that changes in interest rates are the main driver of our findings. The evidence suggests that neither of these last two alternative stories is an important driver of our results.

4.1 Deposit insurance and household portfolio

To explore which assets (if any) were liquidated in response to the changes in deposits, we partnered with the bank and conducted a telephone survey of 990 clients that were randomly selected from our sample. The survey was carried out in November and December of 2019. Despite the fact that all customers were asked the questions, the share of customers who answered the survey was less than 40%, with some variation taking place across questions. This is due to the nature of some questions, which could be perceived as touching on sensible issues. Therefore, while our survey helps us understand how deposit insurance

affects depositors' behavior, its results are suggestive and must be interpreted with caution.

We first asked the clients if they knew about the existence of FOGAFIN, or the deposit insurance scheme, and if they knew about the policy change. Three hundred and eighty-eight clients answered the first question, and 128 answered the second one. Internet Appendix Figure A2 shows the percentage of the different responses. Panel A shows that among those who answered the first question, 32% know about FOGAFIN. Among those who answered the second question, 25% knew about it. These numbers are relatively high and imply that a large fraction of depositors knows about deposit insurance and, in particular, about the policy change.²⁴

To investigate the source of deposit growth, we asked the clients about their deposit growth from 2017 to 2019, and 323 customers responded. Among those with higher deposits, we asked them to specifically indicate the assets that were liquidated and transformed into bank deposits. We offered five possible answers: (1) cash (stored at home or in the office), (2) loan liquidations (someone paid a loan owed to the client), (3) liquidation of financial investments (e.g., stock, bonds, investment funds), (4) sale of fixed assets (e.g., a house, car, or television), and (5) others. This last option corresponds to cases in which the source is different from the liquidation of alternative assets, like receiving an unforeseen income shock.

Twenty-three percent of the clients who answered the question on deposit growth said that they increased deposits, and 40% of these respondents reported that they liquidated some assets as a result. Internet Appendix Figure A3 shows the distribution of liquidated assets, with cash being by far the most important and exceeding a response rate of 50%. Overall, these results suggest that some clients are changing the composition of their savings and lowering their holding of other assets (cash in particular) to increase deposits. This is not necessarily in response to the policy change. But it makes it more likely that individuals who increased their deposits because of the policy change did so using cash stored outside banks.

4.2 Deposit reshuffling and interest rates

Our findings are consistent with a story in which the bank's customers perceive that it is safer to increase deposits above 20 million COP after the policy change and therefore liquidate other assets to increase their deposit holdings. But two other possibilities could explain our results. First, customers with deposit accounts across multiple banks before the policy change could close some of them and consolidate their savings in the account of the bank of this study, leading to an increase in deposits, in particular for individuals in bins 1 and 2. We refer to this possibility as the "reshuffling hypothesis." The ideal test for

²⁴ If more people knew about the policy change we would expect larger effects on deposits. Therefore, we can see our results as a lower bound of the effect of the increase in deposit insurance.

this mechanism requires data from multiple banks, which we lack. Still, in this section, we discuss our previous results in light of this possibility and provide five pieces of additional evidence that suggest that this alternative explanation is not a key driver of our results.

First, we note that two of our previous findings are inconsistent with the reshuffling hypothesis. The event study (Figure 4) shows that the increase in deposits of individuals in bins 1 and 2 is gradual. If reshuffling was an important driver of this increase, we would expect a much faster change in deposits since moving cash across bank accounts is relatively easy, while transforming other less liquid assets to cash takes more time. Second, customers bunching below the insurance threshold across multiple bank accounts are likely to be sophisticated. Therefore, customers who are more likely to reshuffle should have more seniority, and be more likely to have credit products and multiple savings products with the bank. But in Sections 3.5 we showed that these characteristics do not predict a higher response to the change in deposit insurance, in particular for individuals in bins 1 and 2. Under the assumption that these heterogeneity variables are correlated with the propensity to reshuffle, these results are inconsistent with the reshuffling idea.

Now we turn to the discussion of five exercises that provide additional evidence against the reshuffling hypothesis. The first one explores how the effect of deposit insurance varies according to the level of competition faced by each bank branch in the city where it operates. For each branch, we gathered information on the number of competing banks in the city. Individuals with multiple bank accounts (who are more likely to reshuffle) are likely to be more concentrated in places with many banks. Therefore, if deposit reshuffling was an important driver of our results, we would expect the change in deposits to be higher in bank branches where there is high competition.²⁵ Columns 4 and 5 of Table 3 show how the effect of the policy change differs depending on two measures of competition: the number of competing banks (column 4) and a dummy equal to one if the number of competing bank is in the top quartile of the distribution. The estimated coefficient of the interaction term shows that the effect of the policy change does not vary according to the level of competition. This is true in particular for bins 1 and 2. This result is inconsistent with the hypothesis of systematic reshuffling.

Our second exercise investigates the distribution of deposit growth rates in our sample in the period after the policy change. We plot the distribution of deposit growth rates in Internet Appendix Figure A4. Panel A shows the distribution in the range $[-1, 1]$, and panel B zooms in on the range $[-0.2, 0.2]$. If the reshuffling of resources were important in our sample, we would observe bunching at certain points of the distribution. For example, if a large fraction of clients were moving resources from this bank to another, we would expect

²⁵ In Colombia the opening and closing of most savings products require that the borrower goes in person to the bank. Nevertheless, online banking lessens the relevance of geography and therefore of this test.

a large fraction of observations close to a -1 rate. Also, if resources were reshuffled from other banks to this bank, then we would observe a large fraction of observations with high deposit growth rates. In the figure, we observe that most observations fall in the zero bin and the distribution is smooth. At the same time, a sizable mass of depositors presents a small increase in deposits.²⁶ Overall, this descriptive evidence does not support a reshuffling hypothesis, in which an individual consolidates her accounts into a single one and records very large and sudden increases in deposits.

Our third and fourth exercises explore the number of accounts that opened or closed during our period of analysis. Under the reshuffling explanation, we expect a discontinuous change in this number after the policy change. Internet Appendix Figure A5 depicts the percentage of new and closed accounts as a percentage of the total.²⁷ There are two interesting results. First, the bank is expanding, as the share of new accounts exceeds the share of closing accounts in all months (around 0.3% vs. 0.1%). Second, account openings and closings remain relatively stable around the time of the policy change. These results are not consistent with the reshuffling hypothesis. Furthermore, Internet Appendix Figure A6 shows the distribution of the amounts deposited in new accounts that are opened each month from January 2017 to September 2019. If large amounts of resources were being reshuffled, we would expect some bunching around the 50 million COP threshold after the policy change. We would also expect a large fraction of the new accounts in the 20 million to 50 million COP interval. The histograms show that this is not the case.

Finally, we consider the effect of the policy change on those individuals with deposits in a range of 5 million to 15 million COP. Recall that these individuals are not included in the sample of our other analysis, since they are not bunching and they are completely insured both before and after the policy change. Nevertheless, if depositors are splitting their savings across banks, individuals with small amounts in one bank might withdraw these funds to deposit them in another bank, thus consolidating their funds in a single place. If this behavior is important, we would expect individuals with deposits in the range of 5 million to 15 million COP to actually decrease their deposits after the policy change. Internet Appendix Figure A7 shows the result of estimating the event study specification of Equation (2) but using as treatment group individuals in the range of 5 million to 15 million COP. The control group is the same as before, consisting of individuals in bin 3. We find that relative to these depositors, individuals in the range of 5 million to 15 million COP were

²⁶ Note also that this picture rejects evidence of bunching around round numbers (like 50% or 100%).

²⁷ Recall that our main sample is a monthly panel collapsed at the individual level. For this exercise, we use data at the account level (the same individual can have multiple accounts) and say that an account opened at t if the account was not observed before t and in t deposits were at least 5 million COP. We say that an account closing occurs in month t if the account is observed in the data at t , but not at $t+1$, and if it had at least 5 million COP in t .

not on parallel trends before the policy and, if anything, they increased their deposits during the period of analysis.

In sum, none of these tests completely rules out the possibility that deposit reshuffling is happening in our context. But taken together, they suggest that reshuffling is not the sole driver of our results. The second alternative hypothesis that we consider is that sharp changes in interest rates around the time of the policy change could lead to an increase in deposits. Internet Appendix Figure A8 shows the evolution around the policy change of the annual interest rate of 90-day fixed-term deposits. Panel A shows the nominal rate, while panel B shows the same rate but relative to the policy rate of the Banco de la República, the central bank of Colombia. The data come from the same institution. In line with other countries in the world, deposit rates in Colombia declined steadily in 2017, and there was no sharp change around the time of the policy change. Nevertheless, the relative rate increased steadily as shown in panel B. Still, there was no sharp increase in this rate around May 2017, suggesting that changes in the interest rate of fixed-term deposits are not driving our results. If this was the case, then we should not observe parallel trends across bins since the rates were following similar trajectories before and after the policy. The same argument can be made regarding the evolution of the interest rate of savings accounts, which is depicted in Internet Appendix Figure A9.

Overall, the results presented in this section offer evidence against the reshuffling explanation. Changes in interest rates do not seem to drive the results either. Together with the results from the previous sections of the paper, this suggests that depositors are substituting other assets in favor of deposits in safer bank accounts.

5. The Elasticity of Deposits to Deposit Insurance

In this section we estimate the elasticity of deposit growth to deposit insurance. To our knowledge, this parameter has not yet been estimated with the level of precision that our data allows but is key to understanding depositor behavior and calibrating the optimal coverage of deposit insurance like in Davila and Goldstein (2021). We address the following question: how much do deposits increase in response to an increase in insurance coverage? Our estimate relies on the fact that different individuals face a different shock to the share of insured deposits depending on the level of deposits preceding the policy change. To benchmark our result, we provide estimates of this elasticity for the United States.

To perform the analysis, we collapse the data at the individual level i and define the following outcomes. First, $\log(\overline{deposits}_{i,pre})$ is the logarithm of average monthly deposits for individual i in the period before the policy change that we consider in the event studies and DiD of Section 3

(November 2016 to April 2017). $\log(\overline{deposits}_{i,post})$ is defined analogously but for the period after the policy change (May 2017 to November 2017). $\overline{depo_growth}_{i,pre}$ is the average monthly deposit growth before the policy change, and $\overline{depo_growth}_{i,post}$ is defined analogously for the post-period.²⁸ Finally, we define $\Delta Deposits_i = \log(\overline{deposits}_{i,post}) - \log(\overline{deposits}_{i,pre})$ and $\Delta Growth_i = \overline{depo_growth}_{i,post} - \overline{depo_growth}_{i,pre}$. Therefore, for a given depositor i , $\Delta Deposits_i$ measures how the log of average deposits changed between the pre and post-periods. $\Delta Growth_i$ measures how the average growth rate of deposits changed.

We are interested in the relation between $\Delta Deposits_i$ or $\Delta Growth_i$ and the change in deposit insurance between the two periods. Note that before the policy change, the proportion of deposits that was insured was (on average) equal to $(T_{pre} / \overline{deposits}_{i,pre})$, where T_{pre} is equal to 20 million COP. This proportion can be smaller or larger than one depending on the value of $\overline{deposits}_{i,pre}$. Similarly, after the policy change, the proportion of deposits that is insured is $(T_{post} / \overline{deposits}_{i,post})$, where T_{post} is equal to 50 million COP. We define the change in insurance coverage as the difference between these two terms; that is,

$$\Delta Coverage_i = (T_{post} - T_{pre}) / \overline{deposits}_{i,pre}. \quad (5)$$

Consider the following simple example to illustrate the logic of $\Delta Coverage_i$. Suppose that two individuals a and b had average monthly deposits in the preperiod equal to $\overline{deposits}_{a,pre} = 20$ million and $\overline{deposits}_{b,pre} = 100$ million COP. Before the policy change, individual a had a fraction of 1 of her deposits covered, whereas b only had 0.2. After the policy change, a has a fraction of insured deposits of 2.5, and b has a fraction of 0.5. Therefore, $\Delta Coverage_a$ is equal to 1.5, which indicates that the fraction covered for individual a increased by 1.5 points. For individual b , this fraction increased by 0.3 points so that $\Delta Coverage_b = 0.3$.

We are interested in studying the change of the log of deposits for each point that $\Delta Coverage_i$ increases, in other words, the elasticity of deposits to deposit insurance. To obtain this elasticity, we estimate by OLS the simple cross-sectional regression:

$$x_i = \beta \Delta Coverage_i + \varepsilon_i, \quad (6)$$

where x_i is either $\Delta Deposits_i$ or $\Delta Growth_i$. In the first case, β captures the change in log deposits from the pre- to the post-period in response to an increase of one point in the fraction of insurance coverage, and it can

²⁸ Recall from above that deposit growth in t is defined as the change in log deposits; that is, $\log_deposits_{it} - \log_deposits_{it-1}$.

Table 4
Estimates of the elasticity of deposits to deposit insurance

A. Colombia		
	$\Delta Deposits_i$ (1)	$\Delta Growth_i$ (2)
$\Delta Coverage_i$	0.023*** (0.002)	0.019*** (0.001)
Obs.	38,539	38,539
Adj. R^2	0.004	0.026
B. United States		
	Deposit growth (1)	Dom. deposit growth (2)
$Insured_b \times Policy_t$	0.0339** (0.0148)	0.0323** (0.0163)
Bank holding com. FE	Yes	Yes
Quarter FE	Yes	Yes
Obs.	19,768	19,768
Adj. R^2	.183	.166

This table reports estimates of the elasticity of deposits to deposit insurance for Colombia (panel A) and for the United States (panel B). In panel A, we follow the specification described by Equation (6). The dependent variables are at the depositor i level. $\Delta Deposits_i$ is the difference between the log of average deposits in the preperiod (April 2016–April 2017) and the post-period (May 2017–March 2018), and $\Delta Growth_i$ is the difference between the average growth rate in the pre- and post-period. These outcomes are regressed over $\Delta Coverage_i$, defined as the ratio between the increment in deposit insurance, 30 million COP, and the average level of deposits in the preperiod, as described in detail in Section 5. The sample includes depositors in bins 1, 2, 3, and 4. Standard errors are reported in parentheses. In panel B, we present estimates of Equation (8) in a quarterly panel of banking holding companies (BHCs) in the United States. The source of all variables is the Call Reports (FDIC SDI), and the time period is 2008-q1 to 2008-q4. The dependent variable of column 1 is deposit growth, defined as the log change in deposits from quarter $t - 1$ to quarter t . Deposits are given by the BHCs' level of deposits at the end of the quarter. The outcome for column 2 is the growth of domestic deposits. The variable of interest is the interaction between a dummy equal to one from 2008-q3 onward, and zero otherwise, and the share of insured deposits, $Insured_b$. This share is calculated at the BHC level and is equal to the ratio of average insured deposits in the four quarters before 2008-q3 and average total deposits in the same period. Standard errors clustered at the BHC level are reported in parentheses. * $p < .10$; ** $p < .05$; *** $p < .01$.

therefore be interpreted as the elasticity of deposits to deposit insurance. The interpretation when x_i is $\Delta Growth_i$ is similar since in this case β captures the change in average deposit growth from the pre- to the post-period for each point of change in deposit coverage. We report the results of estimating Equation (6) in panel A of Table 4. Column 1 shows that an increase in insurance coverage of 1 point leads to a 2.3-percentage-point increase in the level of deposits. The same increase in insurance leads to an increase of 1.9 percentage points in average deposit growth (column 2).

To benchmark these results, we estimate the magnitude of the elasticity of deposits to deposit insurance in the United States. In this country, the FDIC is the agency that looks after the stability of the financial system through deposit insurance, like FOGAFIN. Since its creation in 1933, the FDIC has been an essential part of the American financial system and regulatory framework. We use quarterly data from depository institutions and exploit the change in the insurance threshold of 2008, triggered by the financial crisis, that extended the

coverage of insured deposits from US\$100,000 to US\$250,000.²⁹ Although this change was motivated by financial distress, we believe these results are useful to benchmark our findings in Colombia.

We use data covering a 1 year window around the 2008 policy change (2008-q1 to 2008-q4) to follow our exercise for Colombia. We focus on the Statistics on Depository Institutions (SDI), which contains the financial information of every FDIC-insured institution at a quarterly frequency. The data include the level of deposits and the fraction insured as well as other characteristics of the institution. The primary source of this information is the Federal Financial Institution Examination Council (FFIEC) Call Reports. Therefore, the SDI data include both data traditionally associated with the FDIC, such as deposit information, and Call Reports data.

The SDI data are available at the institution-quarter-year level, and we collapse it at the level of the bank holding company (BHC) and the quarter-year level.³⁰ We are interested in the total deposits and the total domestic deposits of the BHC, which vary across quarters. We investigate if banks that had a higher share of insured deposits before 2008 recorded a stronger deposit growth after the policy change. Several changes in deposit insurance occurred in the United States before 2008, but we cannot consider them since the last one was implemented in 1980, before the starting year of the SDI data in 1992.

To estimate the elasticity of deposit growth to deposit insurance, we rely on the fact that banks had different levels of exposure to the increase in the deposit insurance threshold depending on the level of insured deposits before the policy change. We first calculate the share of insured deposits on each bank's balance sheet before the change. We define the share of insured deposits as

$$Insured_b = \frac{ID_{b,pre}}{TD_{b,pre}}, \quad (7)$$

where b denotes the BHC and pre refers to the period before the change. ID_{pre} denotes average insured deposits in the four quarters before the policy change, and TD_{pre} denotes average total deposits in the same period.³¹

²⁹ This policy change was announced in late September 2008 and was implemented by December 2008. Its conclusion, and the return from the US\$250,000 to US\$100,000 threshold, was originally planned to take place in December 2010 but was instead extended indefinitely. For details on the policy and its implementation, see FDIC (2008, 2010).

³⁰ This is done using the BHC or Regulatory Top Holder ID (called *RSSDHCR*), which identifies the BHC. A regulatory top holder is any company that directly or indirectly owns, controls, or has power over 25% of the vote or more of a bank, has a direct holding of the majority of a company's shares, controls the election of a majority of the directors or trustees of a bank, holds direct control over the company, or exercises a controlling influence over the management or the policies of a bank.

³¹ Here, we are making the assumption that the share of insured deposits is a good proxy of the share of insured accounts. Martin, Puri, and Ufieri (2018) report that these two quantities are different in the United States which has implications for the interpretations of our findings, as suggested by Davila and Goldstein (2021). This simplification is due to the aggregate nature of the available bank data from the call reports. With more disaggregated data we could compute both shares but in their absence, we treat each bank as a "representative depositor". Despite this limitation, we believe that this assumption is not excessively constraining in our setting, since we estimate similar elasticities for the United States and Colombia.

We estimate the following equation:

$$Growth_{bt} = \beta Post_t \times Insured_b + \alpha_b + \gamma_t + \varepsilon_{bt}, \quad (8)$$

where b denotes the BHC and t a quarter-year period. The dependent variable $Growth_{bt}$ is either total deposit growth or domestic deposit growth.³² $Post_t$ is a dummy equal to one after the policy change (from 2008-q3 onward). α_b and γ_t denote banking holding company and quarter-year fixed effects, respectively. ε_{bt} denotes the error term, which we cluster at the bank level. β , our coefficient of interest, captures the difference in deposit growth from the pre- to the post-period for each additional point of the share of insured deposits and can therefore be interpreted as an elasticity of deposits to deposit insurance.³³

Panel B of Table 4 presents the estimates of Equation (8) for a 1-year window around the time of the policy change. In line with the results in Colombia, both deposit growth and domestic deposit growth are larger after the increase in deposit insurance for those banks that have a larger fraction of insured deposits. The estimate of β in column 1 implies that banks with a 1 point higher fraction of insured deposits $Insured_b$ have a deposit growth rate that is 3.4% higher after the policy change. We obtain a very similar coefficient if we consider domestic deposits only (column 2). The elasticities of the two countries are close in magnitude, albeit the elasticity in the United States is larger.

Estimating the elasticity in the United States is valuable for two reasons. First, this estimate gives a sense of the external validity of our estimates in Colombia. The mechanisms outlined by our theoretical model should in principle apply to settings other than Colombia. They imply that governments can achieve deposit inflows by raising the deposit insurance threshold, which is typically done in times of financial crises and when the probability of bank default is high. The Colombian setting is ideal to isolate the mechanism outlined by the model and estimating the effect of deposit insurance unbiased by the effect of economic forces that are typical in times of economic turmoil.³⁴ The fact that the elasticities of the two countries are of the same sign and of similar magnitude is reassuring in that our estimates in Colombia are valuable to guide policy in other contexts, and in particular in times of financial crises.

Second, and although the elasticities of the two countries are in the same order of magnitude, the elasticity of the United States is 0.0109 points larger than the Colombian one (that is, 47% larger). This difference is large and

³² We define deposit growth as the difference between log deposits in quarter t and quarter $t-1$. Deposits are measured at the end of the quarter.

³³ When considering the outcome of domestic deposit growth, we use a measure of insured deposits, $Insured_b$, computed using only information on domestic deposits.

³⁴ Recall from Section 3.5 that the differential response across bins is not heterogeneous depending on individual characteristics like having a loan product with the bank, the individual's seniority and the number of other savings products. Therefore our results are not driven by specific groups, which is reassuring in terms of the external validity of our results.

likely policy relevant. For example, the difference between these parameters may affect the calibration of the optimal level of deposit insurance, in line with Davila and Goldstein (2021). The upward bias that we document for the United States may be due to fragile banks having an aggressive attitude to attract deposits in the run-up to the financial crisis, in line with the work of Martin, Puri, and Ufier (2018). If prior to 2008 these banks were both raising more deposits (including insured ones) and responded more aggressively to the increase in the insurance threshold, the estimated elasticity in the United States would be biased upward.

6. Conclusions

In this paper, we study the relationship between deposit insurance and deposit behavior. We start by presenting a stylized model that combines features of models in the banking and the public finance literature and assumes a positive probability of bank default and a limited level of deposit insurance. The model predicts that the distribution of deposits exhibits bunching and that an increase in deposit insurance increases deposit holdings. This response is heterogeneous and stronger for individuals who were bunching before the policy change. The driver of this effect is the increase in the expected return of deposits that induces individuals to substitute other assets to increase savings.

We test the predictions of the model using granular data at the deposit-month level from a large Colombian bank together with an unexpected and large increase in the Colombia deposit insurance threshold. This change took place for reasons exogenous to the health of the Colombian economy: to update the real value of the threshold, which had not been changed since the year 2000. We identify bunchers in the data and assign individuals to bins according to their average monthly deposits before the policy change. Bin 1 includes the bunchers (depositors with average deposits between 15 million and 20 million COP). Bin 2 includes individuals with 20 million to 50 million COP. Individuals in bin 3 have prepolicy deposits between 50 million and 75 million COP, and those in bin 4 have deposits above 75 million. Using within-individual variation across time we show that while depositors across different bins were on parallel trends before the policy change, in the 6 months after April 2017 and relative to individuals in bin 3, depositors in bin 1 increased their level of deposits by 4.5%, those in bin 2 by 3.4% and those in bin 4 did not react. We also find that depositors in bin 1 increased the maturity of their deposits, with a 1.2% higher share of deposits in savings and fixed-term accounts. These results are robust to a battery of robustness tests.

Although the mechanism highlighted by our model is that individuals modify their savings decision to increase deposits when deposit insurance increases, we consider other alternative explanations for the increase in deposits. First, we present results from a survey of 990 clients in our sample that indicate that an important fraction of bank clients knew about deposit insurance and changed their assets holdings to increase deposits. This result is suggestive of a change

in asset composition in response to the policy change. Second, we consider the possibility that our results are driven by the reshuffling of resources across bank accounts and present evidence that is not aligned with this possibility. Changes in interest rates do not seem to be the drivers of our results either. Finally, we leverage the cross-sectional heterogeneity in the exposure to the treatment to estimate an elasticity of deposit growth to deposit insurance of 2.3%. This point estimate is close to the elasticity of 3.4% that we obtain in an exercise exploiting the 2008 threshold increase in the United States, which may be higher due to confounders related to the financial crisis.

All in all, this paper offers important insights into the effects of deposit insurance for various reasons. First, the model offers a simple explanation of how individual depositor behavior changes in response to a change in deposit insurance. Second, we validate the predictions of the model using a policy change that isolates the effect of deposit insurance, unconfounded from the factors that are common in periods of crisis or financial distress. Third, the paper quantifies the elasticity of deposit growth to deposit insurance, which can be useful for theoretical and empirical researchers, policymakers, and bankers. Fourth, our results imply that financial regulation can promote the size and depth of financial systems and credit markets.

References

- Allen, F., E. Carletti, I. Goldstein, and A. Leonello. 2018. Government guarantees and financial stability. *Journal of Economic Theory* 177:518–57.
- Artavanis, N., D. Paravisini, C. Robles-Garcia, A. Seru, and M. Tsoutsoura. 2019. Deposit withdrawals. Working paper. Virginia Tech.
- Atmaca, S., K. Kirschenmann, S. Ongena, and K. J. Schoors. 2020. Deposit insurance, bank ownership and depositor behavior. Working Paper, Faculty of Economics and Business Administration, Ghent University, Belgium.
- Bachas, P., P. Gertler, S. Higgins, and E. Seira. 2021. How debit cards enable the poor to save more. *Journal of Finance* 76:1913–57.
- Breza, E., and C. Kinnan. 2021. Measuring the equilibrium impacts of credit: Evidence from the indian microfinance crisis. *Quarterly Journal of Economics* 136:1447–97.
- Bruche, M., and J. Suarez. 2010. Deposit insurance and money market freezes. *Journal of Monetary Economics* 57:45–61.
- Bryant, J. 1980. A model of reserves, bank runs, and deposit insurance. *Journal of Banking and Finance* 4:335–44.
- Bustos, P., G. Garber, and J. Ponticelli. 2020. Capital accumulation and structural transformation. *Quarterly Journal of Economics* 135:1037–94.
- Calomiris, C. W., and M. Jaremski. 2019. Stealing deposits: Deposit insurance, risk-taking, and the removal of market discipline in early 20th-century banks. *Journal of Finance* 74:711–54.
- Chernykh, L., and R. A. Cole. 2011. Does deposit insurance improve financial intermediation? Evidence from the Russian experiment. *Journal of Banking and Finance* 35:388–402.
- Chetty, R., J. N. Friedman, T. Olsen, and L. Pistaferri. 2011. Adjustment costs, firm responses, and micro vs. macro labor supply elasticities: Evidence from Danish tax records. *Quarterly Journal of Economics* 126:749–804.

- Choudhary, M. A., and N. Limodio. 2021. Liquidity risk and long-term finance: Evidence from a natural experiment. *Review of Economic Studies* 89:1278–1313.
- Cooper, R., and T. W. Ross. 2002. Bank runs: Deposit insurance and capital requirements. *International Economic Review* 43:55–72.
- Davila, E., and I. Goldstein. 2021. Optimal deposit insurance. Working Paper, Yale University.
- Demirguc-Kunt, A., E. Kane, and L. Laeven. 2015. Deposit insurance around the world: A comprehensive analysis and database. *Journal of Financial Stability* 20:155–83.
- Demirguc-Kunt, A., and E. Detragiache. 2002. Does deposit insurance increase banking system stability? An empirical investigation. *Journal of Monetary Economics* 49:1373–406.
- Diamond, D. W., and P. H. Dybvig. 1983. Bank runs, deposit insurance, and liquidity. *Journal of Political Economy* 91:401–19.
- Diamond, D. W., and R. G. Rajan. 2000. A theory of bank capital. *Journal of Finance* 55:2431–65.
- Egan, M., A. Hortaçsu, and G. Matvos. 2017. Deposit competition and financial fragility: Evidence from the US banking sector. *American Economic Review* 107:169–216.
- Fama, E. F., and K. R. French. 1992. The cross-section of expected stock returns. *Journal of Finance* 47:427–65.
- FDIC. 2008. FDIC extends opt-out deadline for participation in the temporary liquidity guarantee program. <https://www.fdic.gov/news/press-releases/2008/pr08110.html>. Accessed: May 1, 2022.
- . 2010. Basic FDIC insurance coverage permanently increased to 250,000 per depositor. <https://www.fdic.gov/news/press-releases/2010/pr10161.html>. Accessed: May 1, 2022.
- FOGAFIN. 2017. Resolución 2 de 2017. https://normativa.colpensiones.gov.co/colpens/docs/pdf/resolucion_fogafin_0002_2017.pdf. Accessed: April 4, 2022.
- . 2022. <https://www.fogafin.gov.co/que-es-el-seguro-dedepositos/definicion>. Accessed: April 28, 2022.
- Gatti, M., and T. Oliviero. 2019. Deposit insurance and banks' deposit rates: Evidence from the 2009 EU policy change. Working Paper, Bicocca Department of Economics, Management and Statistics, University of Milan.
- Goldstein, I., and A. Pauzner. 2005. Demand-deposit contracts and the probability of bank runs. *Journal of Finance* 60:1293–327.
- International Monetary Fund. 2016. Detailed assessment of observance of key attributes of effective resolution regimes for financial institutions – Pilot of the draft assessment methodology. IMF Country Report No.16/99.
- Iyer, R., T. L. Jensen, N. Johannesen, and A. Sheridan. 2016. The run for safety: Financial fragility and deposit insurance. Technical Report, EPRU Working Paper Series.
- Iyer, R., T. Lærkholm Jensen, N. Johannesen, and A. Sheridan. 2019. The distortive effects of too big to fail: Evidence from the Danish market for retail deposits. *Review of Financial Studies* 32:4653–95.
- Iyer, R., and M. Puri. 2012. Understanding bank runs: The importance of depositor-bank relationships and networks. *American Economic Review* 102:1414–45.
- Kleven, H. J. 2016. Bunching. *Annual Review of Economics* 8:435–64.
- La Republica. 2017. Ampliación del seguro de depósitos de FOGAFIN fue solo una actualización. <https://www.larepublica.co/finanzas/ampliacion-del-seguro-de-depositos-de-fogafin-fue-solo-una-actualizacion-castano-2498566>. Accessed: April 7, 2022.
- . 2018. FOGAFIN aumentó la cobertura de seguros de depósito a 50 millones. <https://www.larepublica.co/economia/fogafin-aumento-la-cobertura-delos-seguros-de-20-millones-a-50-millones-2753168>. Accessed: May 1, 2022.
- Martin, C., M. Puri, and A. Ufieri. 2018. Deposit inflows and outflows in failing banks: The role of deposit insurance. Working Paper, National Bureau of Economic Research.

Perez-Reyna, D., and D. Osorio. 2018. The monetary and fiscal history of Colombia: 1960-2017. Working Paper, Becker Friedman Institute for Economics, University of Chicago.

Presidencia de la República. 2017. Aumenta de 20 millones a 50 millones cobertura del seguro de depósitos. <http://es.presidencia.gov.co/noticia/170419-Gobierno-aumenta-de-20-millones-a-50-millones-cobertura-del-seguro-de-depositos>. Accessed: May 1, 2022.

Reinhart, C. M., and P. Wickham. 1994. Commodity prices: Cyclical weakness or secular decline? *IMF Economic Review* 41:175–213.

Repullo, R. 2000. Who should act as lender of last resort? An incomplete contracts model. *Journal of Money, Credit and Banking* 32:580–605.

Revista Semana. 2017. Fogafin aumentó la cobertura de seguros de depósito a 50 millones. <https://www.semana.com/economia/articulo/fogafin-amplia-sucobertura-de-seguros/244269/>. Accessed: April 28, 2022.

Slutzky, P., and M. Chavaz. 2020. Retaining worried depositors: Evidence from multi-brand banks. Working Paper, Bank of England.

van der Kwaak, C., J. Madeira, and N. Palma. 2021. The long-run effects of risk: An equilibrium approach. Discussion Paper, the University of Manchester.